

WatchSchedule Product Roadmap

Scheduling Intelligence & Fairness Platform

Vision

Transform WatchSchedule from a watch rota generator into the industry's first Fairness Operating System™ for superyacht watchkeeping.

The platform should progress through structured development phases, ensuring early commercial value while building increasingly powerful scheduling intelligence and data advantages over time.

Phase 1 – Core Scheduling Platform (MVP)

Objective

Deliver a working watch scheduling platform capable of generating and publishing schedules for operational use onboard.

Features

Crew Management

- Crew database
- Rank management
- Watchkeeper eligibility
- Rotation management
- Relief crew support

Leave Management

- Leave requests
- Leave approval workflow
- Leave visibility within scheduling

Schedule Generation

- Single-watch vessels
- Double-watch vessels
- Triple-watch vessels
- Automatic watch allocation

Schedule Management

- Weekly view
- Monthly view
- Rotation view

- Schedule publication
- Schedule history

Captain Controls

- Manual edits
- Approval workflow
- Schedule locking
- Version history

Success Criteria

- Generate a complete watch schedule in under 30 seconds.
 - Eliminate spreadsheet-based scheduling.
 - Publish bridge-ready schedules.
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Phase 2 – Fairness Engine™

Objective

Introduce measurable fairness into schedule generation.

Features

Crew Fairness Score™

Measures fairness experienced by an individual crew member.

Schedule Fairness Score™

Measures fairness of the schedule as a whole.

Duty Weighting System

Weighted allocations for:

- Standard weekdays
- Fridays
- Weekends
- Public holidays
- Christmas periods
- New Year periods

Historical Fairness Score™

Rolling fairness calculations over time.

Fairness Debt™

Track accumulated scheduling imbalance.

Most Due To Serve™

Identify which crew member should receive priority consideration in future allocations.

Success Criteria

- Fairness metrics visible throughout platform.
 - Fairness incorporated into scheduling decisions.
 - Historical fairness retained indefinitely.
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Phase 3 – Fairness Correction Loop™

Objective

Create a self-correcting scheduling engine.

Features

Historical Learning

Every published schedule updates:

- Crew Fairness Score™
- Fairness Debt™
- Historical Fairness Score™

Fairness Recovery

Future schedules actively improve fairness for disadvantaged crew.

AI Fairness Optimisation

Scheduling engine seeks to:

- Improve lowest Crew Fairness Score™
- Reduce Fairness Debt™
- Improve overall Schedule Fairness Score™

Fairness Recovery Tracking

Monitor fairness improvement over time.

Success Criteria

- Fairness Debt™ decreases over time.

- Lowest fairness scores improve through successive schedules.
 - Demonstrable fairness correction across scheduling periods.
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Phase 4 – Explainability & Decision Support

Objective

Provide transparency into scheduling decisions.

Features

Explain My Schedule™

Crew-level explanation engine.

Examples:

- Why was I allocated this duty?
- Why was I assigned this Friday?

Manual Override Tracking

Compare:

- AI recommendation
- Human changes
- Impact on fairness

Schedule Health Score™

Operational quality metric.

Measures:

- Coverage risks
- Duty clustering
- Excessive overrides
- Resource shortages

Rotation Stability Score™

Measures schedule consistency and predictability.

Success Criteria

- Every allocation can be explained.
 - Captains understand impact of manual changes.
 - Reduced scheduling disputes.
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Phase 5 – Schedule Intelligence Assistant™

Objective

Provide a natural language interface to scheduling data.

Features

Ask the Schedule™

Examples:

- Why did Jack receive this duty?
- Who is most due for a weekend off?
- What happens if Sarah takes leave next month?
- Increase fairness above 90.
- Show highest Fairness Debt™.

What-If Analysis

Simulate changes without affecting live schedules.

Schedule Optimisation Requests

Allow users to request:

- Higher fairness
- Better continuity
- Reduced burden

Historical Queries

Natural language access to scheduling history.

Success Criteria

- Captains can interrogate schedules without manual analysis.
- Reduced time spent reviewing schedules.
- Increased trust in AI-generated allocations.

Phase 6 – Predictive Scheduling Intelligence

Objective

Move from reactive scheduling to proactive planning.

Features

Leave Forecasting

Predict fairness impact of future leave.

Coverage Forecasting

Identify future resource gaps.

Fairness Alerts™

Examples:

- Fairness Score below threshold.
- Excess Friday allocation.
- Significant Fairness Debt™ emerging.

AI Recommendations

Examples:

- Consider assigning next Friday to Jack.
- Consider relief cover in August.
- Fairness can be improved by 6 points.

Success Criteria

- Platform identifies issues before they occur.
 - Captains receive actionable recommendations.
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Phase 7 – Analytics & Reporting

Objective

Provide management-level visibility.

Features

Fairness Dashboard

Crew-level fairness analytics.

Fairness Timeline™

Historical fairness trends.

Operational Reporting

Monthly reports:

- Fairness Score
- Duty allocation
- Holiday distribution
- Fairness trends

Owner / Management Reports

Fleet-wide visibility.

Success Criteria

- Management receives measurable fairness metrics.
 - Improved transparency across operations.
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Phase 8 – Fleet Intelligence Platform

Objective

Create a defensible data moat.

Features

Fleet Benchmarking™

Compare vessel performance against fleet averages.

Examples:

- Fairness Score
- Rotation Stability
- Duty distribution

Cross-Vessel Analytics

Identify operational trends.

Fleet Scheduling Insights

Benchmark scheduling quality across vessels.

Success Criteria

- Proprietary fleet-level fairness dataset.
- Industry-leading scheduling intelligence.
- Strong competitive barrier to entry.

Strategic Prioritisation

Tier 1 – Core Commercial Product

Must Build

- Schedule Generation
- Crew Management
- Leave Management
- Crew Fairness Score™
- Schedule Fairness Score™
- Fairness Debt™
- Historical Fairness Score™
- Most Due To Serve™

Tier 2 – Competitive Advantage

Build After MVP

- Fairness Correction Loop™
- Explain My Schedule™
- Manual Override Tracking
- Schedule Health Score™
- Rotation Stability Score™

Tier 3 – Premium Intelligence Layer

Premium Features

- Ask the Schedule™
- What-If Simulator
- AI Recommendations
- Fairness Alerts™
- Leave Forecasting

Tier 4 – Long-Term Data Moat

Enterprise Features

- Fleet Benchmarking™
- Fleet Analytics
- Owner Reporting
- Cross-Vessel Intelligence

End State Vision

Most scheduling platforms generate rotas.

WatchSchedule continuously learns from historical scheduling decisions, measures fairness, corrects imbalance over time, explains every allocation and provides AI-driven operational intelligence.

The longer a vessel uses the platform, the smarter, fairer and more valuable it becomes.

This creates a powerful long-term competitive advantage that cannot easily be replicated by traditional scheduling software.